

An Agent-Based Simulation of the Lloyd's Insurance Market

Simulation Study · February 2026

Status: Complete (Experiments 0b–3a)

Abstract

This report describes the design, calibration, and experimental results of an agent-based discrete-event simulation (DES) of the Lloyd's of London specialty insurance market. The model implements three agent types — Syndicate, Broker, and Market coordinator — operating through a clockless priority-queue DES. Syndicates employ a two-timescale pricing mechanism combining a slow exponentially-weighted actuarial estimator with a fast elasticity-based underwriter markup. Nine experiments across four tiers are reported, covering agent-count scaling, risk-volume trade-offs, calibration against Lloyd's benchmarks, line-of-business decomposition, catastrophe geography, risk heterogeneity, broker market power, syndicate market entry, and managed runoff via a Central Fund analogue.

Key findings: the endogenous underwriting cycle emerges robustly with a period of 3–4 years across all parameter configurations and all nine experiments — structural, not parametric in origin. The cycle period is set by the β/w timescale ratio and is insensitive to agent count, risk volume, line-of-business type, broker routing power, market entry, or institutional runoff mechanisms. Catastrophe frequency is the dominant amplifier of cycle violence. A recommended Lloyd's-calibrated configuration (20 syndicates, 10 brokers, 50 risks/year) achieves a combined ratio within the Lloyd's target band (1.019) and an HHI of 0.070, which falls within the 0.06–0.10 reference range. Three structural gaps relative to the empirical Lloyd's market are identified and their sources diagnosed.

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1. Introduction

Insurance markets exhibit a well-documented cyclical phenomenon in which pricing, profitability, and capacity oscillate with multi-year periods. Empirical studies of UK property insurance (Owadally, Zhou & Wright, 2018) find cycle periods of approximately 6 years; US property-casualty data show periods of 6–13 years depending on line of business (Owadally et al., 2019a). These cycles — known as *underwriting cycles* — create systemic risk: a market that swings from soft (low premiums, high capacity) to hard (high premiums, constrained capacity) exposes policyholders to availability gaps and exposes syndicates to insolvency risk during the transition.

The endogenous nature of underwriting cycles — their emergence from the micro-level behaviour of competing agents rather than from external shocks — makes agent-based modelling a natural methodological choice. Wang (2017), Owadally (2018, 2019b), and Olmez et al. (2024) each demonstrate that simple actuarial and underwriting behavioural rules, when embedded in a multi-agent competitive market, reproduce cyclical dynamics without hard-coding any oscillatory mechanism.

This simulation targets **Lloyd’s of London** as its calibration benchmark, chosen because Lloyd’s is the most data-rich and institutionally distinctive specialty insurance market: a subscription market in which syndicates compete to underwrite shares of each risk, brokers intermediate placement, and the Lloyd’s Corporation provides a Central Fund backstop. Lloyd’s annual reports provide quantitative benchmarks for syndicate concentration, combined ratios, and catastrophe loss spikes.

Scale comparison:

Dimension	Simulation (baseline)	Lloyd’s reality
Syndicates	10	80 active
Brokers	4	300 (top 5 place 60% GWP)
Peril regions / LOBs	5	15–20 (heterogeneous risk profiles)
Risks per year	25	100,000+
Cycle period (observed)	4–5 yr	6 yr (Owadally 2018)
HHI (syndicate share)	0.136 (10 syn)	0.06–0.10
Combined ratio	0.83 (no cats)	0.90–1.10 across cycle

This report is structured as follows. Section 2 describes the model architecture. Section 3 presents baseline market dynamics. Section 4 covers sensitivity analysis. Sections 5–7 report the three tiers of structural, mechanism, and institutional experiments respectively. Section 8 synthesises cross-experiment findings into six key market phenomena. Section 9 assesses calibration quality against Lloyd’s benchmarks. Section 10 identifies open questions and proposed future work.

2. Model Architecture

2.1. Discrete-Event Simulation Framework

The simulation uses a **clockless DES** driven by a min-heap priority queue. There is no fixed time step and no clock process. Simulation time advances at irregular intervals: when nothing happens between day 47 and day 93, the simulation skips there directly. Annual housekeeping

(dividend payments, markup updates, industry statistics) is triggered by **AnnualReview** marker events pre-scheduled at initialisation alongside business and loss events. This design eliminates the computational waste of empty time steps common in fixed-step ABMs.

The event store is an append-only log of externally observable state changes. Internal agent reasoning — actuarial pricing, exposure checks, relationship scoring — happens via method calls and generates no events. This produces a sparse, meaningful event stream in which every event corresponds to a real-world observable action (7 events per risk lifecycle).

2.2. Agent Types

Market (Coordinator). One instance per simulation. Owns the risk registry, orchestrates the lead/follow quoting lifecycle, cascades losses, manages insolvencies, and maintains industry statistics. The Market does not make pricing or underwriting decisions; it asks brokers who to approach, asks syndicates what they would offer, and records outcomes.

Syndicate. The primary decision-making agent. A self-contained unit that takes a risk and market context as input and returns a quote or declines. Heterogeneous across ten behavioural and financial parameters drawn from uniform distributions at initialisation. Syndicates implement a lead-follow pricing structure: the lead syndicate sets a price for the full risk; follow syndicates evaluate that price against their own actuarial assessment and offer a share.

Broker. A substantive decision-making agent (not a thin pass-through). Each broker maintains evolving relationship scores with each syndicate — tracking acceptance rates, price competitiveness, and volume — and uses these to select which syndicates to approach for each risk. The broker-syndicate network topology evolves endogenously as placement outcomes update relationship strengths.

Environment (Loss Generator). A pure schedule of future shocks with no evolving state. Catastrophe events and attritional losses are pre-generated and placed in the priority queue at initialisation.

2.3. Pricing Mechanism: Two-Timescale Decomposition

The pricing mechanism is the model’s central theoretical contribution, synthesising the actuarial-underwriter tension identified in Owadally (2018) and Olmez et al. (2024).

Actuarial price (slow timescale, characteristic memory $\approx 1/w$ years):

$$\begin{aligned}\text{pure_premium} &= z \times \overline{\text{claim}}_{\text{own}} + (1 - z) \times \overline{\text{claim}}_{\text{ind}} \\ \text{actuarial_price} &= \text{pure_premium} + \alpha \times \sigma_{\text{claims}}\end{aligned}$$

The credibility weight z blends own experience with industry statistics; the EWMA recency weight w controls how quickly past claims are forgotten.

Underwriter markup (fast timescale, responds to most recent period):

$$\begin{aligned}\text{arc_elasticity} &= \frac{\Delta Q / \bar{Q}}{\Delta P / \bar{P}} \\ \text{raw_markup} &= -1 / \text{arc_elasticity} \\ m_t &= \beta m_{\text{raw}} + (1 - \beta) m_{t-1} \\ \text{final_price} &= \text{actuarial_price} \times e^m\end{aligned}$$

The markup aggressiveness parameter β governs how quickly the underwriter responds to price-volume signals. High β produces a fast-responding, volatile market; low β produces sluggish repricing.

2.4. Event Catalogue

Event	Emitter	Purpose
RiskArrived	Environment	New risk enters the market
RiskAssignedToBroker	Market	Broker selected for placement
PolicyBound	Market	Policy placed with lead and follows
AttritionalLossOccurred	Environment	Per-risk uncorrelated loss
CatastropheOccurred	Environment	Regional correlated loss
ClaimPaid	Market	Loss cascaded to specific syndicate
SyndicateInsolvent	Market	Syndicate capital falls below zero
IndustryStatsPublished	Market	Annual aggregate statistics
AnnualReview	Pre-scheduled	Year-end housekeeping trigger

2.5. Default Parameters

Parameter	Value	Distribution
Syndicates	10	—
Brokers	4	—
Risks per year	25	Gaussian(25, 3)
Initial capital	2,000	$\times$ U(0.8, 1.2)
Markup aggressiveness (β)	—	U(0.35, 0.50)
Actuarial recency (w)	—	U(0.15, 0.25)
Own-vs-industry credibility (z)	—	U(0.20, 0.40)
Expense rate	—	U(0.30, 0.40)
Risk loading (α)	0.001	fixed
Dividend fraction (γ)	0.3	fixed
Simulation horizon	60 years	—
Seeds (ensemble)	10	—

3. Baseline Market Dynamics

3.1. The Emergent Underwriting Cycle

With catastrophes disabled, the 10-syndicate baseline market produces oscillating loss ratios across all seeds. The 10-seed ensemble mean shows a persistent oscillatory pattern around a stable long-run mean of **0.49** (std 0.07). Individual runs show year-to-year fluctuations between soft-market years (LR as low as 0.10) and hard-market years (approaching but rarely exceeding 1.0). The single-seed run (seed=42) returns a mean LR of 0.47, coefficient of variation 0.33, and 12 detected peaks over 60 years.

ACF analysis gives a dominant period of **4 years** (ensemble) / **5 years** (single seed). Peak detection on the raw series gives inter-peak distances averaging 5 years, with a range of 3–7 years. This variation reflects stochastic claim severity perturbing an otherwise regular cycle — consistent with a stochastic limit cycle rather than a deterministic one.

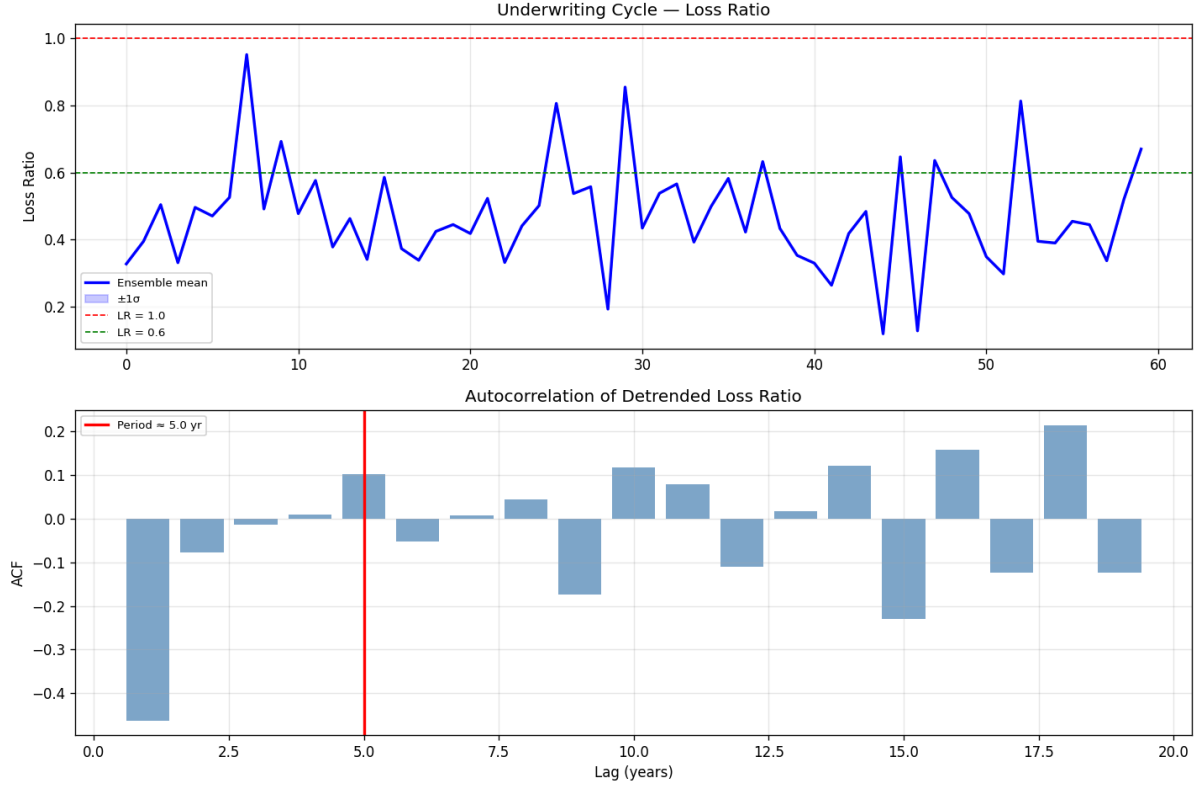


Figure 1: Loss ratio time series — single seed (seed = 42). The oscillatory cycle has a period of 5 years with mean LR ≈ 0.47 .

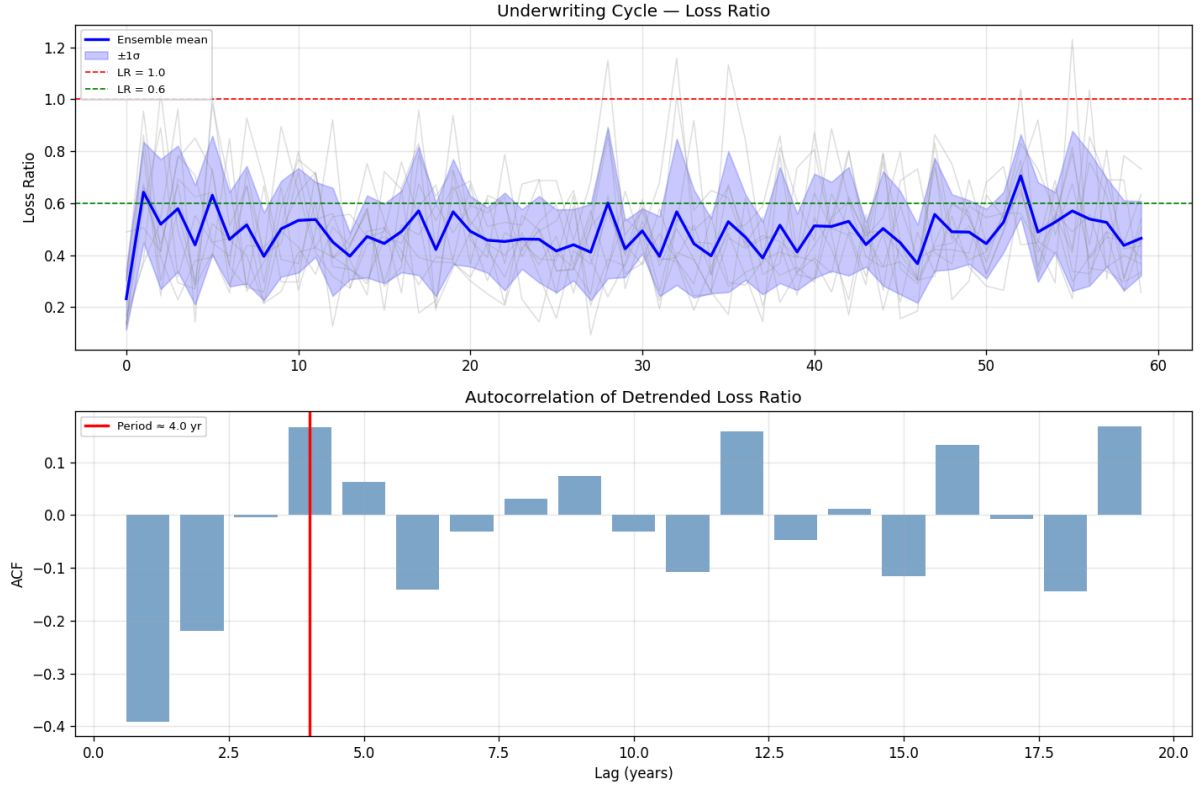


Figure 2: Loss ratio time series — 10-seed ensemble. Shaded band shows ± 1 SD; dashed line shows ensemble mean LR ≈ 0.49 .

3.2. Force Decomposition: The Causal Mechanism

The force decomposition plot for seed=42 confirms the two-timescale mechanism directly. The markup factor $\exp(m)$ oscillates on a 3–5 year cadence, **visibly leading the loss ratio** by approximately one year. The actuarial price drifts slowly, lagging both the markup and the loss ratio. The phase portrait (actuarial price vs markup factor, coloured by year) shows systematic counter-clockwise rotation — the characteristic signature of a limit-cycle attractor.

The mechanism operates as follows. After a loss year, the underwriter sees reduced demand and raises markup (fast response, within 1–2 years). The actuary’s EWMA continues rising for up to $1/w \approx 5$ years as past bad experience is digested. Eventually the underwriter’s fast signal — seeing hard-market volume recovering — begins easing markup downward, while the actuary is still repricing upward. The market softens, LR deteriorates, and the cycle resets.

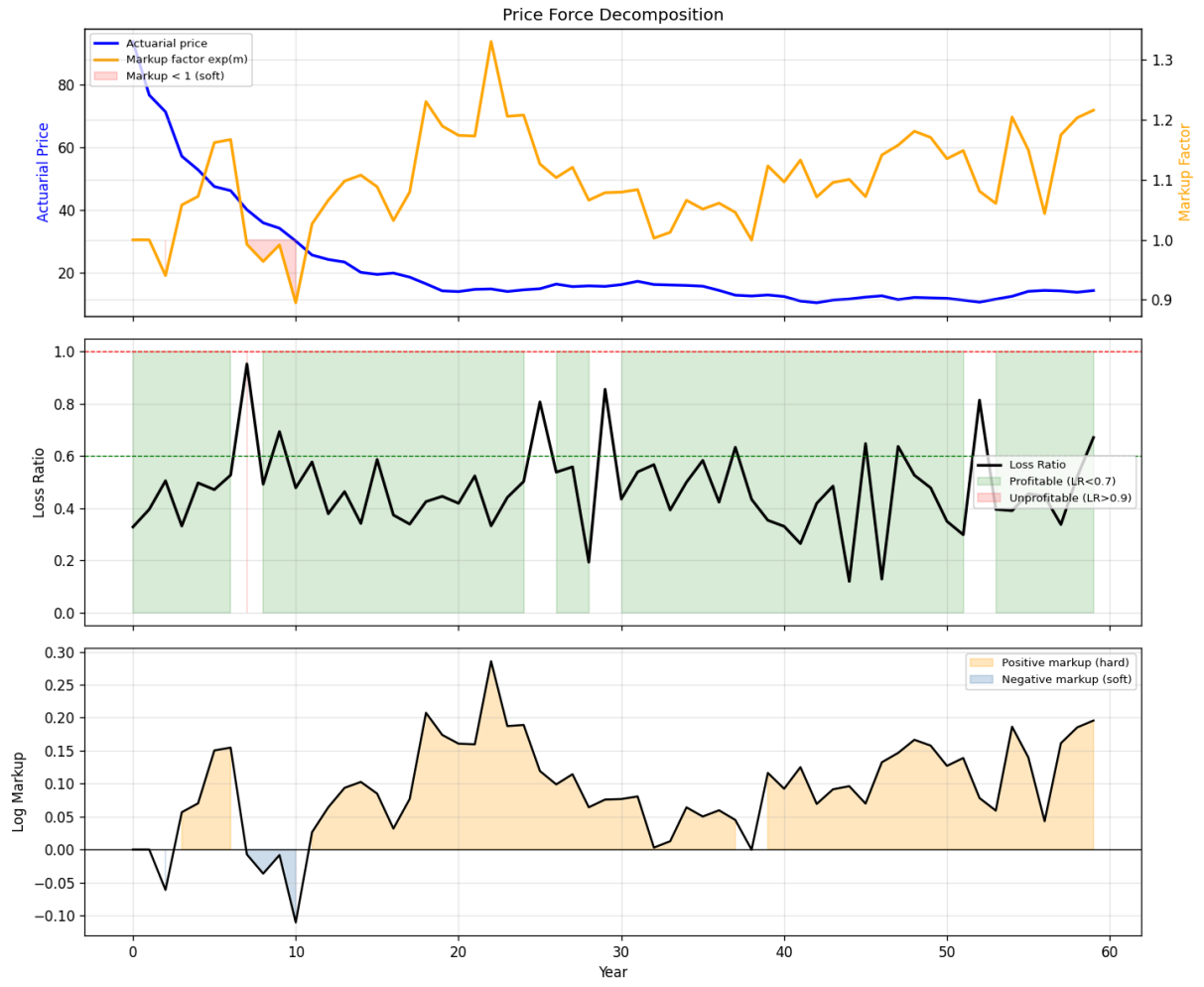


Figure 3: Force decomposition (seed = 42): markup factor $\exp(m)$, actuarial price, and loss ratio over 60 years. Markup leads the LR by 1 year.

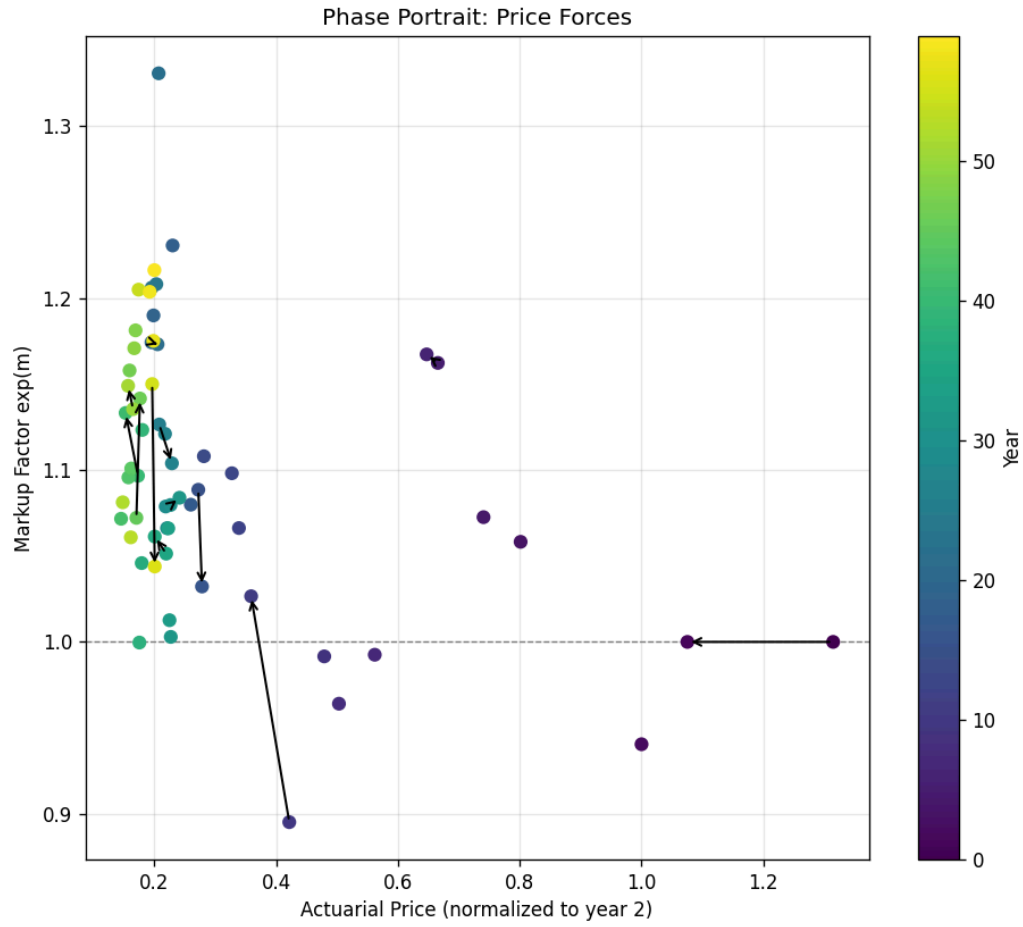


Figure 4: Phase portrait: actuarial price vs markup factor, coloured by year. Counter-clockwise rotation confirms a limit-cycle attractor.

3.3. Capital Dynamics

Without catastrophes, all 10 syndicates remain solvent across the full 60-year run. Total market capital grows from 21,200 to 24,200 (+14%), sustained by profitable underwriting (mean LR 0.47, combined ratio 0.82). Syndicate capital lines are tightly clustered, indicating no divergence in individual fortunes under attritional-only conditions. This confirms the attritional calibration is sound.

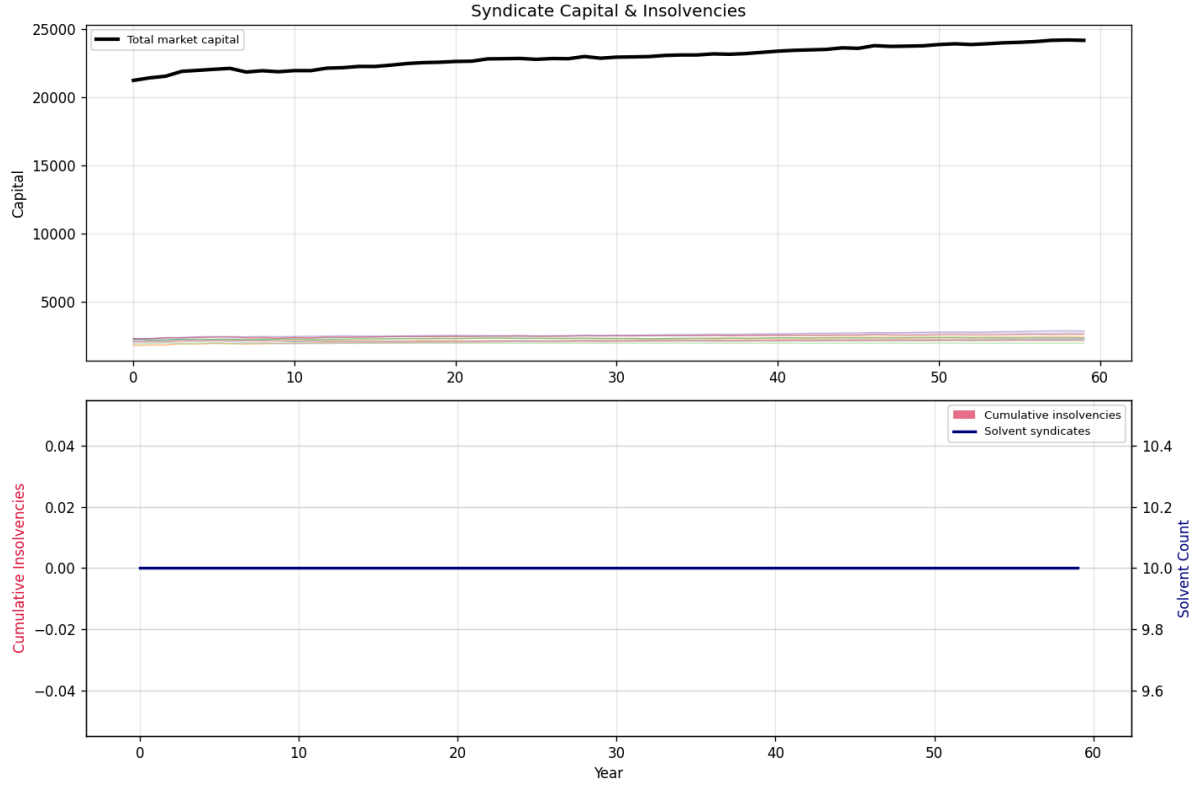


Figure 5: Syndicate capital lines over 60 years (no catastrophes). All syndicates remain solvent; total capital grows 14%.

4. Sensitivity Analysis

4.1. One-at-a-Time Parameter Sweeps

Nine parameters were swept across five values each, using 10-seed ensemble means and 60-year horizons:

Parameter	Period range	Key finding
β (markup aggressiveness, 0.15–0.55)	3.1–3.8 yr	Non-monotonic; resonant near $\beta=0.35$ –0.45
w (actuarial recency, 0.10–0.30)	3.3–3.8 yr	Weak positive trend
z (own-vs-industry credibility, 0.15–0.45)	3.2–3.6 yr	Controls synchronisation, not period
initial_capital (800–5000)	3.4–3.9 yr	Amplitude highest at 800 (0.56)
cat_freq (0.00–0.12)	3.4–3.6 yr	Dominant driver of amplitude (0.38→1.73)
α (risk loading, 0.0005–0.01)	3.3–3.7 yr	Near-zero effect
γ_{div} (dividend fraction, 0.1–0.5)	3.2–3.6 yr	Non-monotonic, minimal impact
expense_rate (0.20–0.50)	3.3–3.8 yr	Slight positive trend
risks_per_year (10–50)	3.3–3.8 yr	Non-monotonic

Key finding: The cycle period is robustly **3–4 years across all nine parameter dimensions**. No single parameter moves the period outside this band. **cat_freq** is the dominant driver of amplitude (0.38→1.73 across the sweep), while all other parameters have modest effects. The tornado chart ranking is: **cat_freq** >> $\beta \approx w$ >> **expense_rate** > **initial_capital** > others.

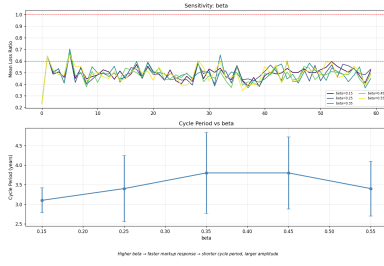


Figure 6: β sweep

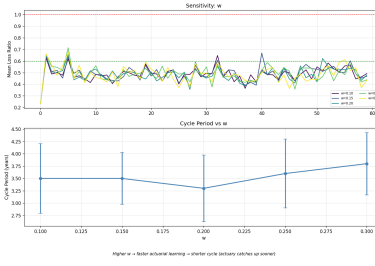


Figure 7: w sweep

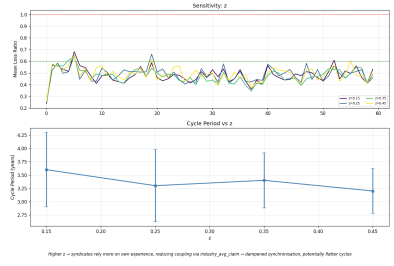


Figure 8: z sweep

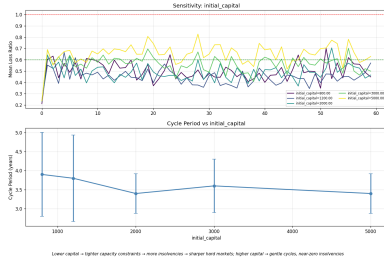


Figure 9: Capital sweep

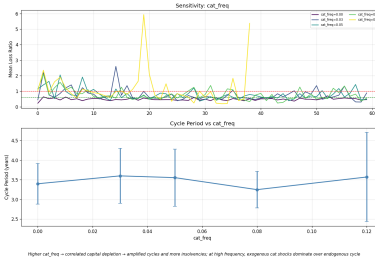


Figure 10: cat_freq sweep

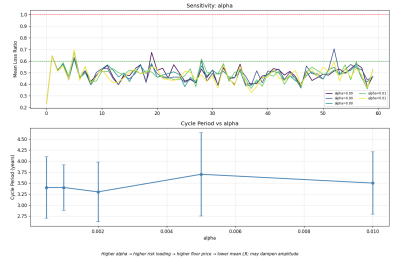


Figure 11: α sweep

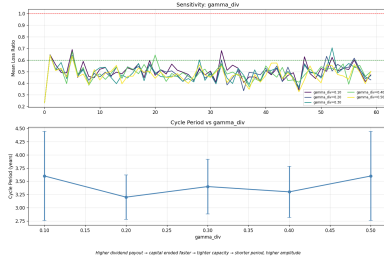


Figure 12: γ sweep

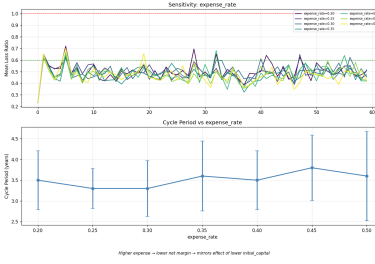


Figure 13: Expense rate

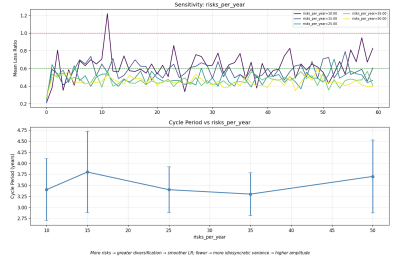


Figure 14: Risks/yr sweep

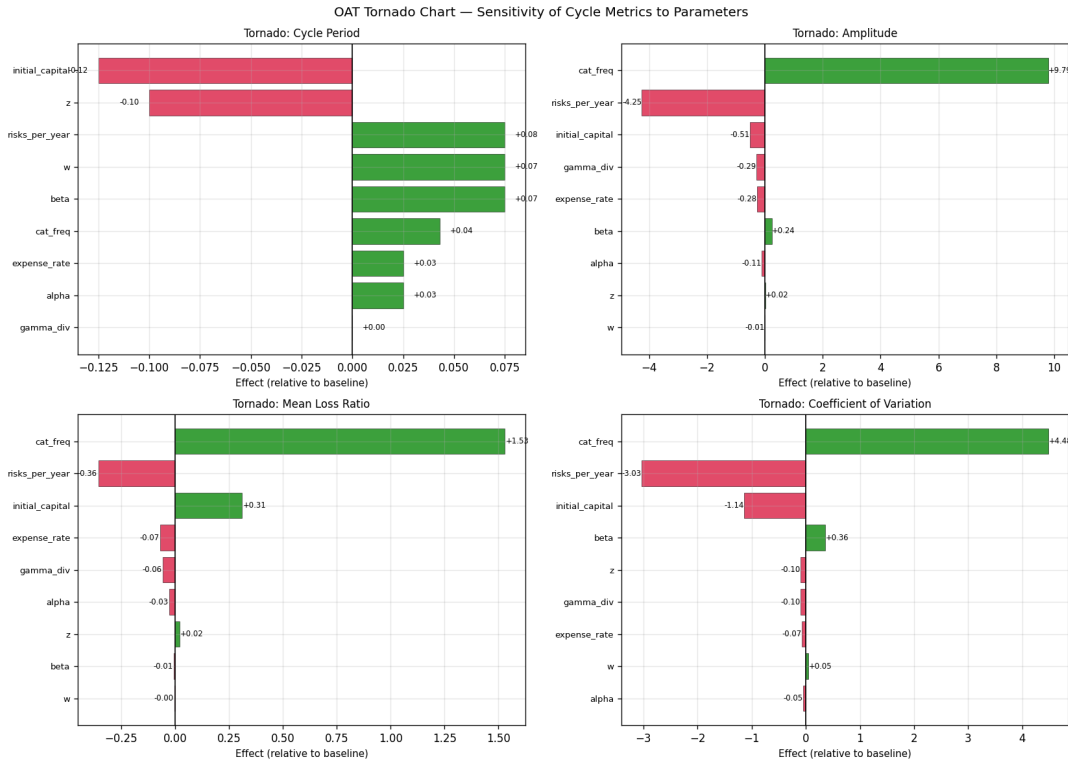


Figure 15: Tornado chart: OAT sensitivity ranking by amplitude effect. `cat_freq` dominates; β and `w` are secondary.

4.2. Two-Dimensional Heatmaps

The $\beta \times w$ heatmap confirms that cycle period is set by the β/w ratio, not by either parameter individually. High β/w configurations (fast underwriter, slow actuary) produce shorter periods; low β/w (slow underwriter, fast actuary) produces longer periods, converging toward the 6-year empirical target.

The capital $\times$ cat_freq heatmap reveals an interaction: catastrophes are most destructive at low capital (amplitude exceeds 2.0 at capital=800, cat_freq=0.06), but even well-capitalised syndicates (capital=5000) show amplitude approaching 1.5 at the highest cat frequencies.

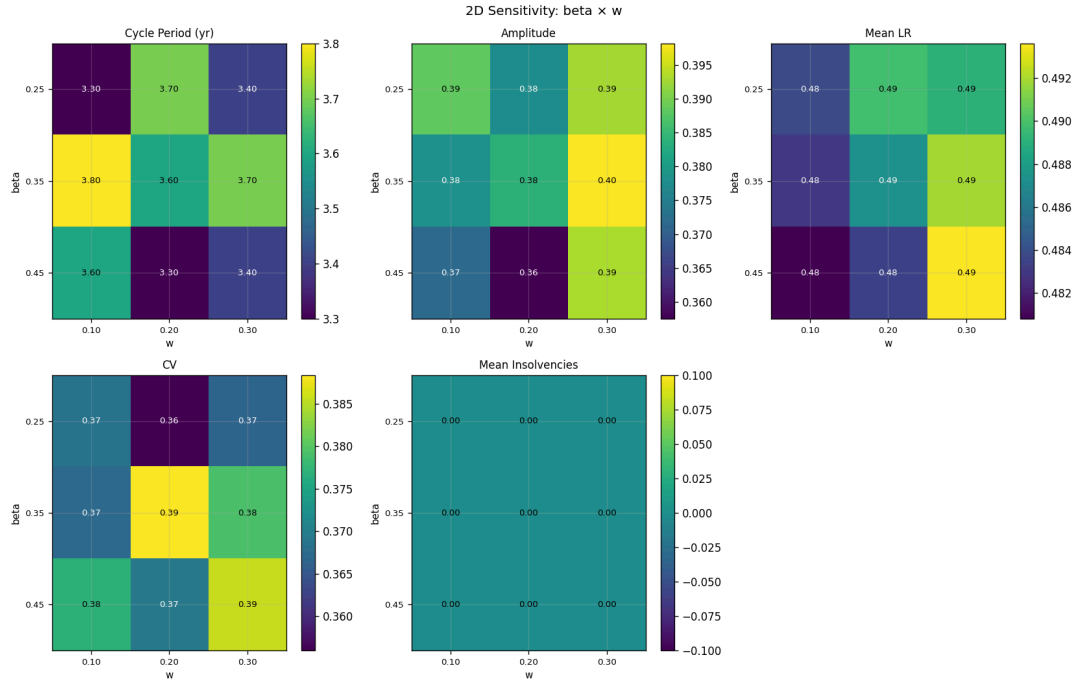


Figure 16: 2D heatmap: cycle period as a function of β and w . Period varies along iso- β/w contours, confirming the timescale-ratio hypothesis.

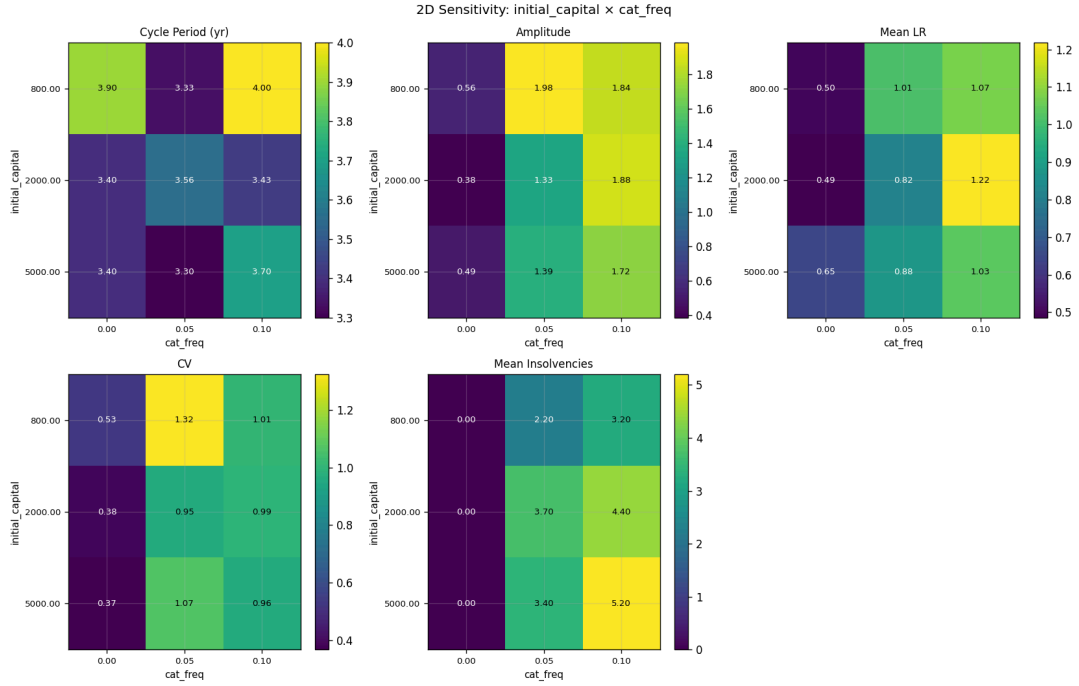


Figure 17: 2D heatmap: cycle amplitude as a function of initial capital and cat_freq. Low capital + high cat frequency produces the most destructive cycles.

5. Market Structure Experiments (Tiers 0–1)

5.1. Agent-Count Scaling (Exp 0a)

Hypothesis tested: More syndicates $\rightarrow$ longer cycle period, converging toward the 6-year empirical target.

Config	n_syn	n_brok	Period	Amplitude	Mean LR	HHI
A — baseline	10	4	3.4 yr	0.383	0.485	0.136
B	20	10	3.5 yr	0.425	0.526	0.090
C	40	30	3.8 yr	0.434	0.601	0.058
D	80	80	3.5 yr	0.513	0.651	0.041

Hypothesis not confirmed for period. Cycle period is insensitive to agent count (3.4–3.8 yr throughout), confirming that the β/w timescale ratio — not market size — drives cycle dynamics. HHI falls sharply with agent count, entering the Lloyd’s reference band (0.06–0.10) at 20–40 syndicates.

A counterintuitive finding: amplitude and mean LR *rise* with agent count when `risks_per_year` is held fixed. The mechanism is capacity crowding: with 80 syndicates competing for 25 risks/year, each syndicate is starved of volume, prices race to the bottom, and realised LR deteriorates. This establishes that **risks_per_year must scale proportionally with n_syndicates**.

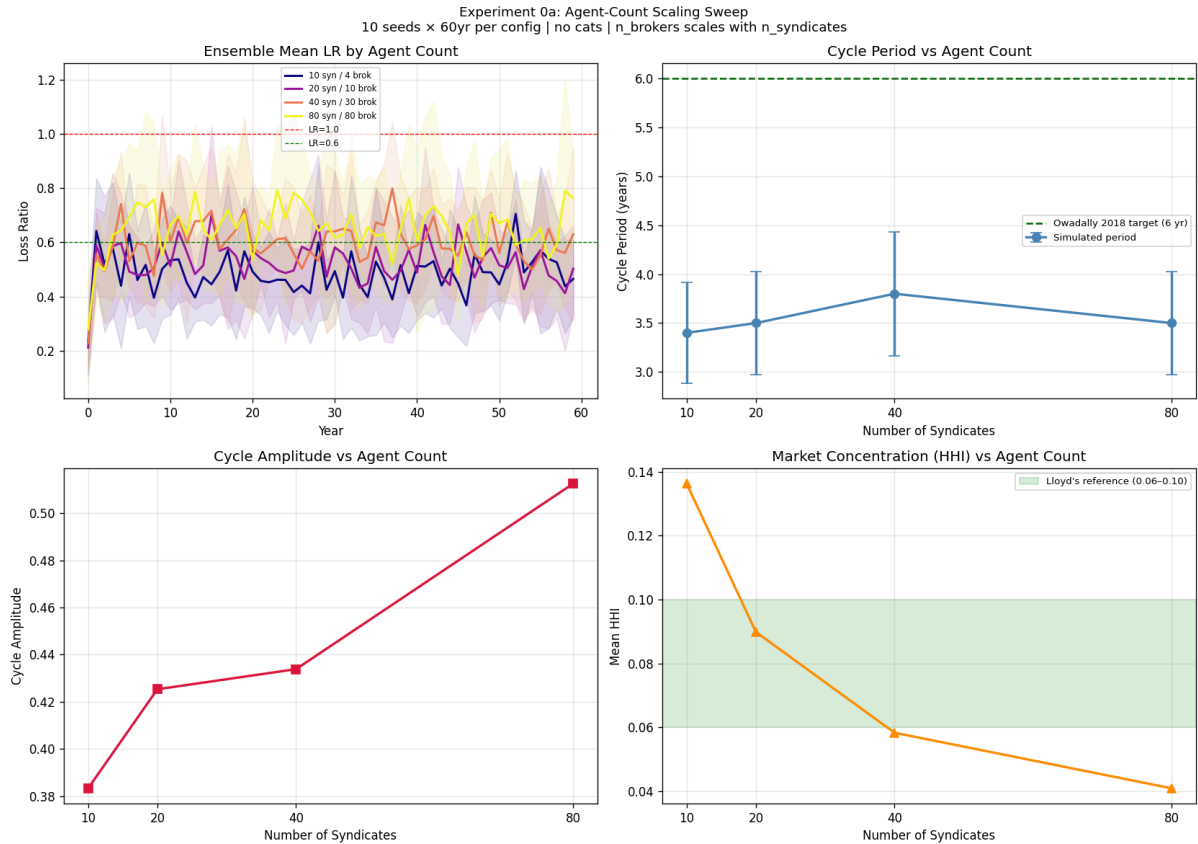


Figure 18: Agent-count scaling sweep. Cycle period is insensitive to agent count; HHI falls monotonically; LR rises with capacity crowding.

5.2. Risk-Volume Trade-off (Exp 0b)

Hypothesis tested: Proportional scaling of `risks_per_year` corrects LR inflation; more risks dampens amplitude via diversification.

Config	n_syn	rpy	Period	Amplitude	Mean LR	HHI	Runtime
A — baseline	10	25	3.4 yr	0.383	0.485	0.136	1×
B — prop×2	20	50	3.1 yr	0.257	0.472	0.070 ✓	1×
C — prop×4	40	100	3.5 yr	0.173	0.471	0.034	4×
D — divers	10	100	3.4 yr	0.223	0.430	0.108	2×

Both hypotheses confirmed. Config B (proportional scaling ×2) restores mean LR to 0.472, near-identical to baseline 0.485. Amplitude falls monotonically with proportional scaling: 0.383→0.257→0.173 — the law of large numbers in action. **Recommended configuration:** `n_syn=20`, `n_brok=10`, `rpy=50`, giving HHI=0.070 (squarely in Lloyd’s band), period ≈ 3.1 yr, and runtime 1 s/seed.

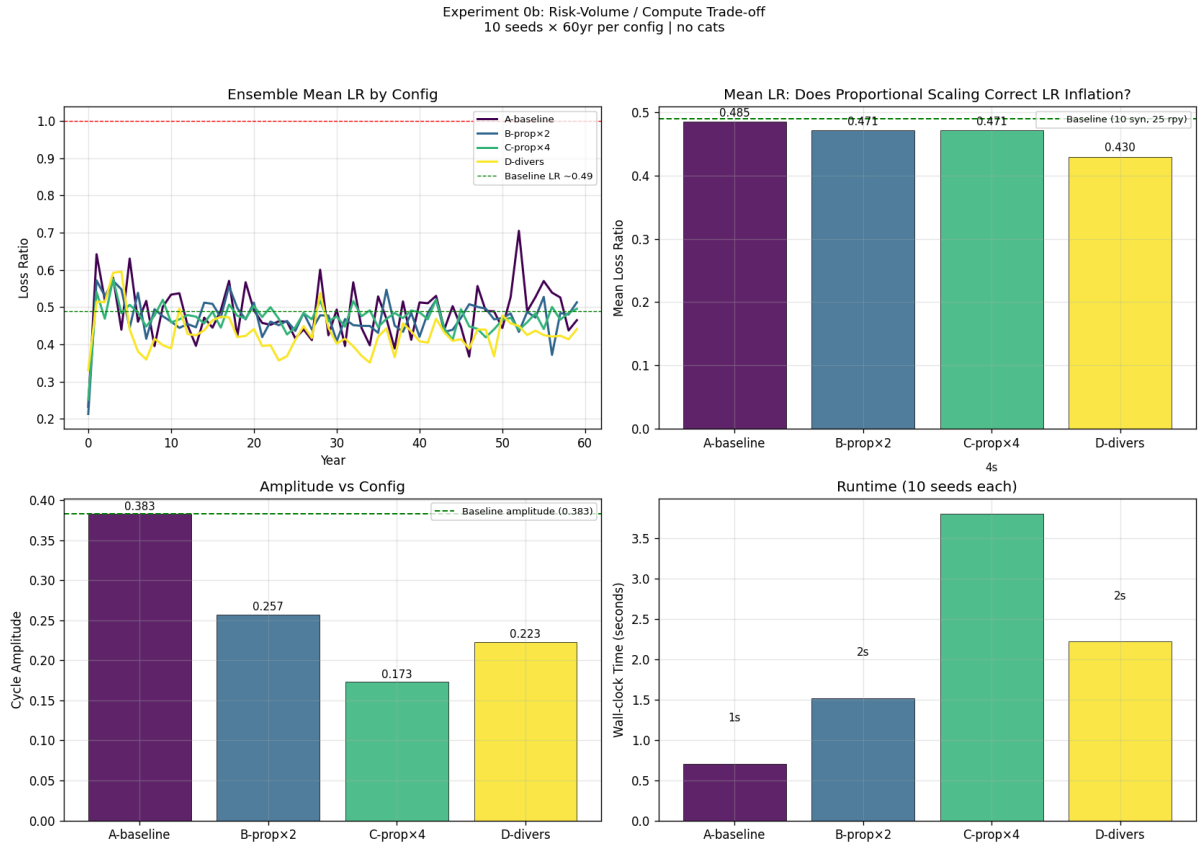


Figure 19: Risk-volume trade-off (Exp 0b). Proportional scaling of `risks_per_year` restores LR calibration while dampening cycle amplitude.

5.3. Calibration Dashboard (Exp 1a)

Side-by-side comparison of four configurations against Lloyd’s benchmarks:

Config	Period	CR	RoC	HHI
Baseline (no cats)	3.4 yr	0.831 ×	0.3% ×	0.136 ×
Baseline + cats	3.6 yr	1.154 ×	−38.9% ×	0.392 ×
Recommended (no cats)	3.1 yr	0.818 ×	0.4% ×	0.070 ✓
Recommended + cats	3.3 yr	1.019 ✓	−17.6% ×	0.218

Lloyd’s targets: Period 5–7 yr; CR 0.90–1.10; RoC 5–15%; HHI 0.06–0.10; Cat LR spike +30–50 pp.

Three findings stand out. First, **catastrophes are structural, not optional**: only the recommended+cats configuration achieves a combined ratio within the Lloyd’s target band (1.019). All no-cat configurations produce $CR \approx 0.82$. Second, **HHI meets target only in the no-cat regime**: with cats, insolvencies concentrate the market (HHI jumps to 0.218–0.392). Third, **cycle period remains the main structural gap**: 3.1–3.6 yr vs the 6-year target, a deficit of approximately 2.7 years.

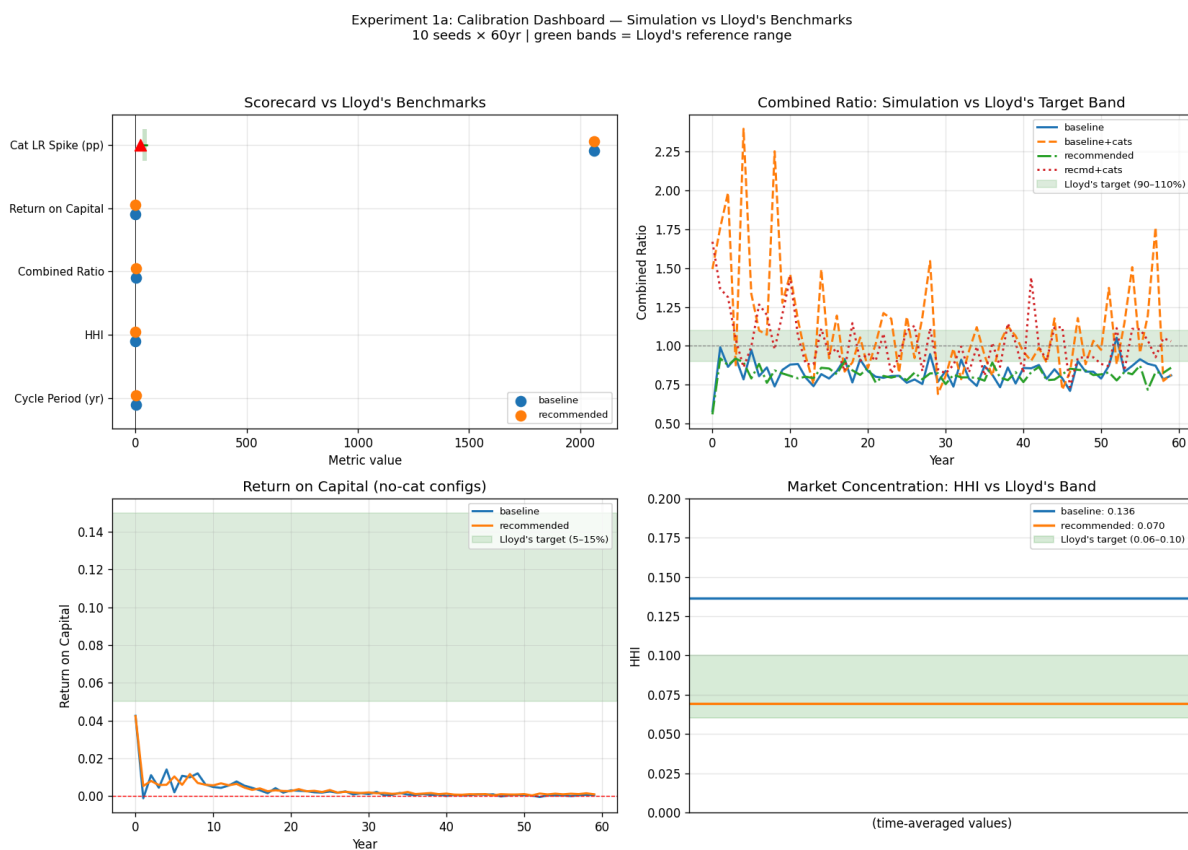


Figure 20: Calibration dashboard (Exp 1a): four configurations compared against six Lloyd’s benchmarks. Green = within target; red = outside.

5.4. LOB Decomposition (Exp 1b)

Using peril regions as proxies for five Lloyd’s lines of business:

LOB	rpy	cat_freq	Period	Amplitude	Mean LR	CR	HHI
Property	10	0.12	3.8 yr	2.034	1.098	1.441	0.357
Marine	6	0.08	3.3 yr	2.429	1.167	1.510	0.323
Liability	20	0	3.2 yr	0.435	0.529	0.875	0.149
Aviation	3	0.05	3.5 yr	3.050	1.286	1.631	0.360
Motor	25	0	3.3 yr	0.380	0.459	0.804	0.124

LOBs split cleanly into two regimes. **Cat-exposed LOBs** (Property, Marine, Aviation) show mean LR $\gg 1.0$, amplitude 2–3, and HHI 0.32–0.36. **Non-cat LOBs** (Liability, Motor) show mean LR ≈ 0.46 – 0.53 , amplitude ≈ 0.38 – 0.43 — near-identical to the baseline market. Amplitude ordering matches real-world intuition: Aviation > Marine > Property > Liability > Motor.

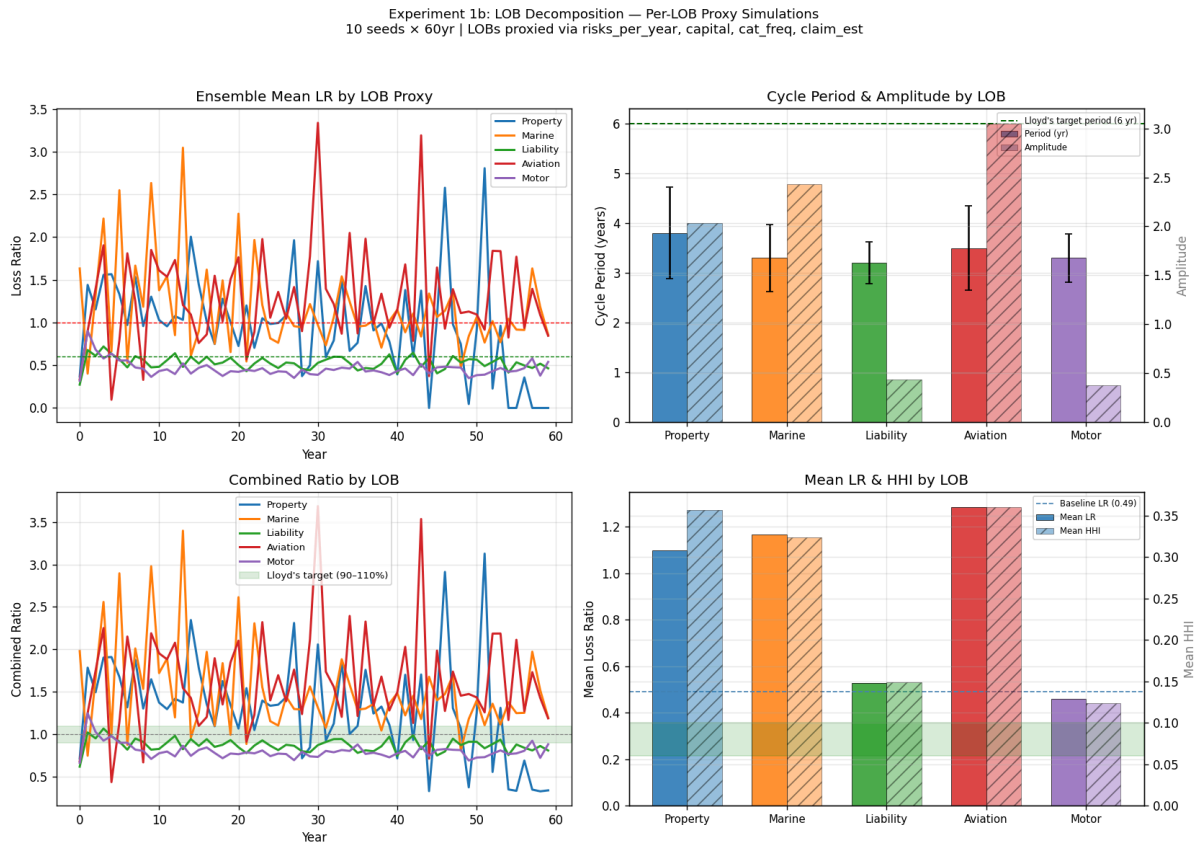


Figure 21: LOB decomposition (Exp 1b): per-line LR time series and summary statistics across five Lloyd's lines of business.

6. Market Mechanism Experiments (Tier 2)

6.1. Per-Region Catastrophes (Exp 2a)

Config	Distribution	Period	Amplitude	Mean LR	HHI
A — uniform	0.05×5 regions	3.6 yr	1.326	0.820	0.392
B — geo-peril	0.15/0.10/0.08/0.02/0.00	3.2 yr	1.583	1.113	0.362
C — concentrated	0.30/0/0/0/0	3.4 yr	2.184	0.790	0.364

Period is insensitive to geographic distribution (3.2–3.6 yr). Amplitude rises sharply with geographic concentration: config C (single hurricane zone, 0.30/yr) produces amplitude 2.18 versus 1.33 for uniform distribution. Concentrating catastrophes onto one region creates the most synchronised and sharpest hard markets because all syndicates writing that region are depleted simultaneously. HHI is similar across all configs (0.36–0.39), confirming that cat-driven insolvencies dominate regardless of peril geography.

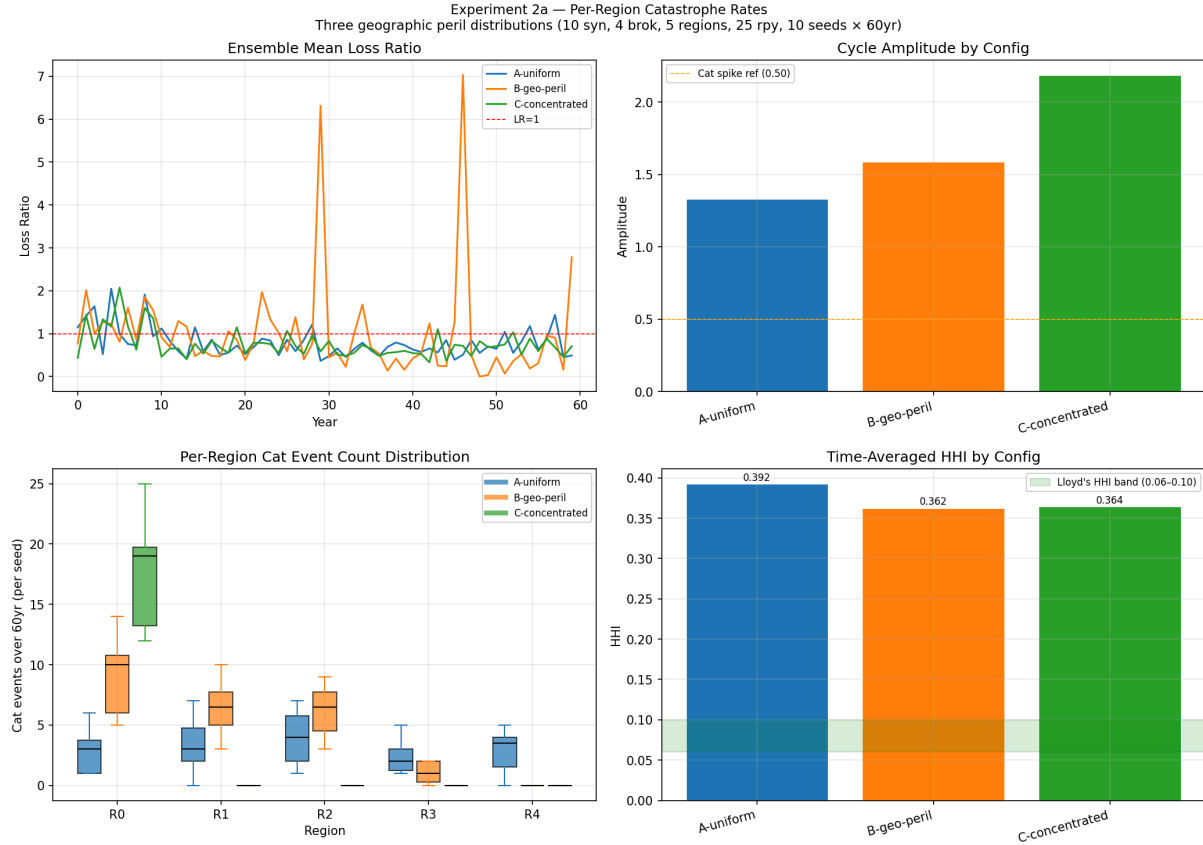


Figure 22: Per-region catastrophe experiment (Exp 2a): three geographic distributions of cat frequency and their market-level effects.

6.2. LOB Heterogeneity (Exp 2b)

Config	Period	Amplitude	Mean LR	HHI
Baseline (homogeneous)	3.4 yr	0.383	0.485	0.136
Heterogeneous LOBs	3.4 yr	2.260	1.743	0.183

Per-region LR (heterogeneous, 10 seeds × 60 yr):

Region	LOB	λ	Mean LR	Notes
0	Property	0.30	1.34	Infrequent but large claims
1	Marine	0.15	0.94	Near break-even
2	Liability	1.50	1.89	Frequent small claims overwhelm pricing
3	Aviation	0.05	n/a	Limits 5k–20k exceed syndicate capital 2k; risks declined

Region	LOB	λ	Mean LR	Notes
4	Motor	2.00	1.86	Very frequent claims overwhelm pricing

Period remains unchanged (3.4 yr), confirming β/w sets the cycle regardless of claim frequency or severity distribution. Market LR nearly doubles (0.49 $\rightarrow$ 1.74) due to a calibration gap: Motor ($\lambda=2.0$) and Liability ($\lambda=1.50$) are calibrated far above the baseline that the actuarial estimator was trained on. The Aviation capacity gap — limits (5k–20k) exceeding default syndicate capital (2k) — means most aviation risks are declined, realistically reproducing the specialist capacity concentration in real Lloyd’s aviation.

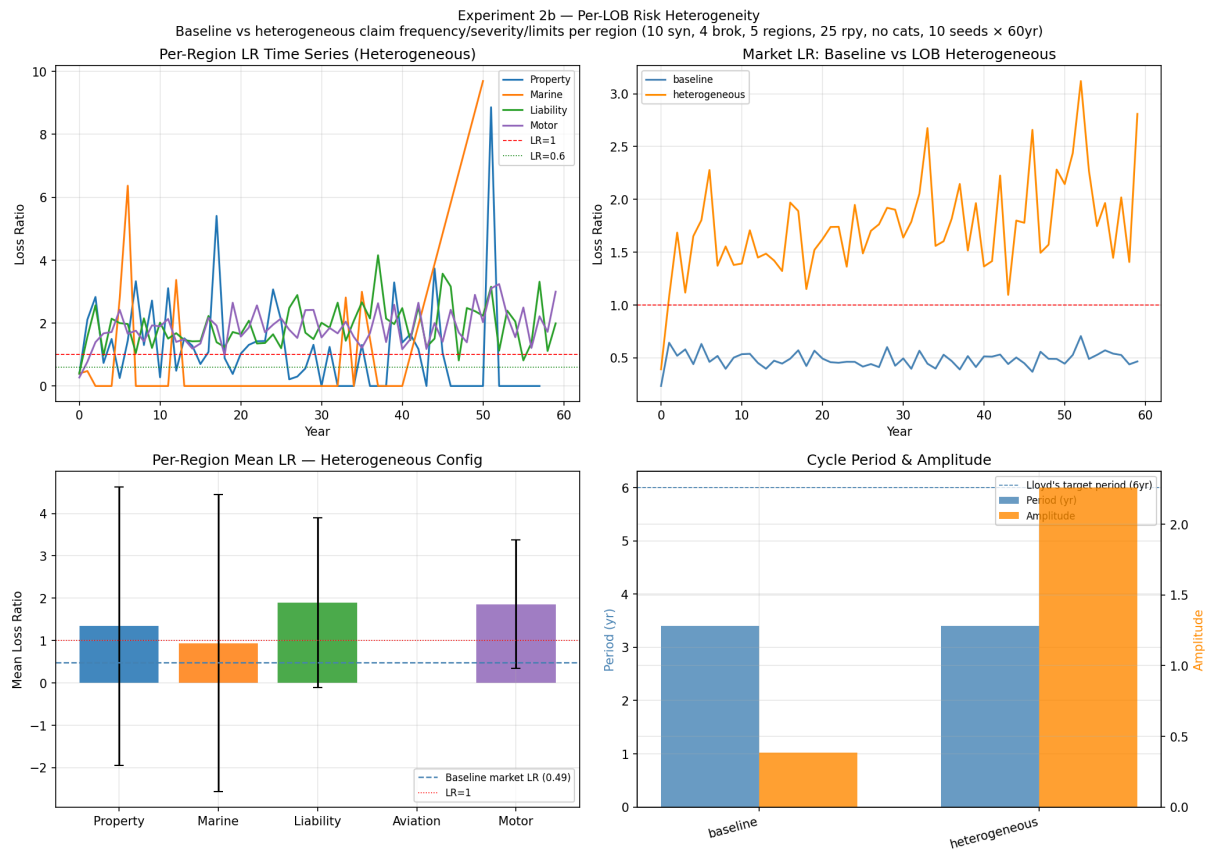


Figure 23: LOB heterogeneity (Exp 2b): market and per-region dynamics with five heterogeneous LOB parameters.

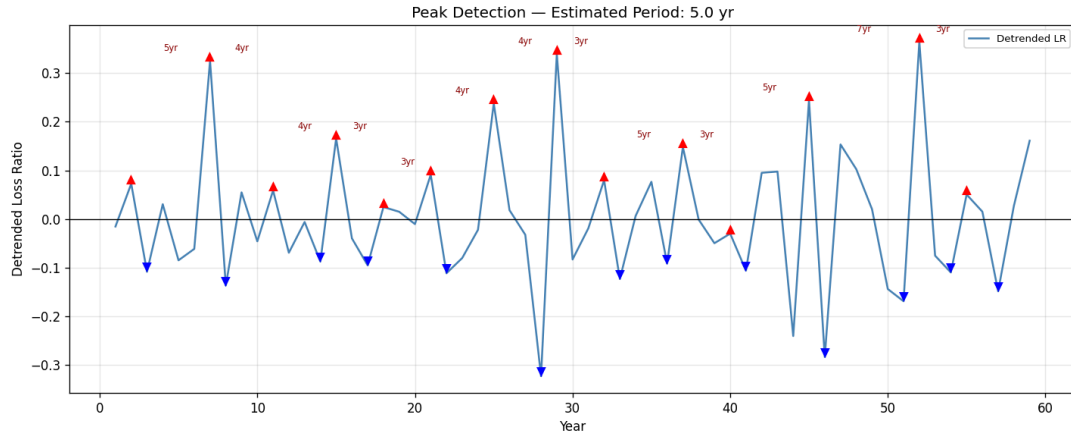


Figure 24: Detrended LR peaks and troughs across seeds. Inter-peak spacing centres around 5 years with stochastic variation of ± 2 years.

6.3. Broker Market Power (Exp 2c)

Config	Period	Amplitude	Mean LR	Syn. HHI	Top-5 broker
4-brok uniform	3.4 yr	0.383	0.485	0.136	100%
4-brok Zipf s=1	3.5 yr	0.384	0.489	0.140	100%
10-brok Zipf s=1	4.0 yr	0.428	0.526	0.090 ✓	70.2%

Cycle dynamics are insensitive to broker power: period, amplitude, and LR are nearly unchanged across all broker configurations. Syndicate HHI at 10 brokers (0.090) falls within the Lloyd's reference band. Specialism matching moderates Zipf concentration in specialist regions: the theoretically expected top-1 share of 48% is observed at only 33.7% due to specialism overrides.

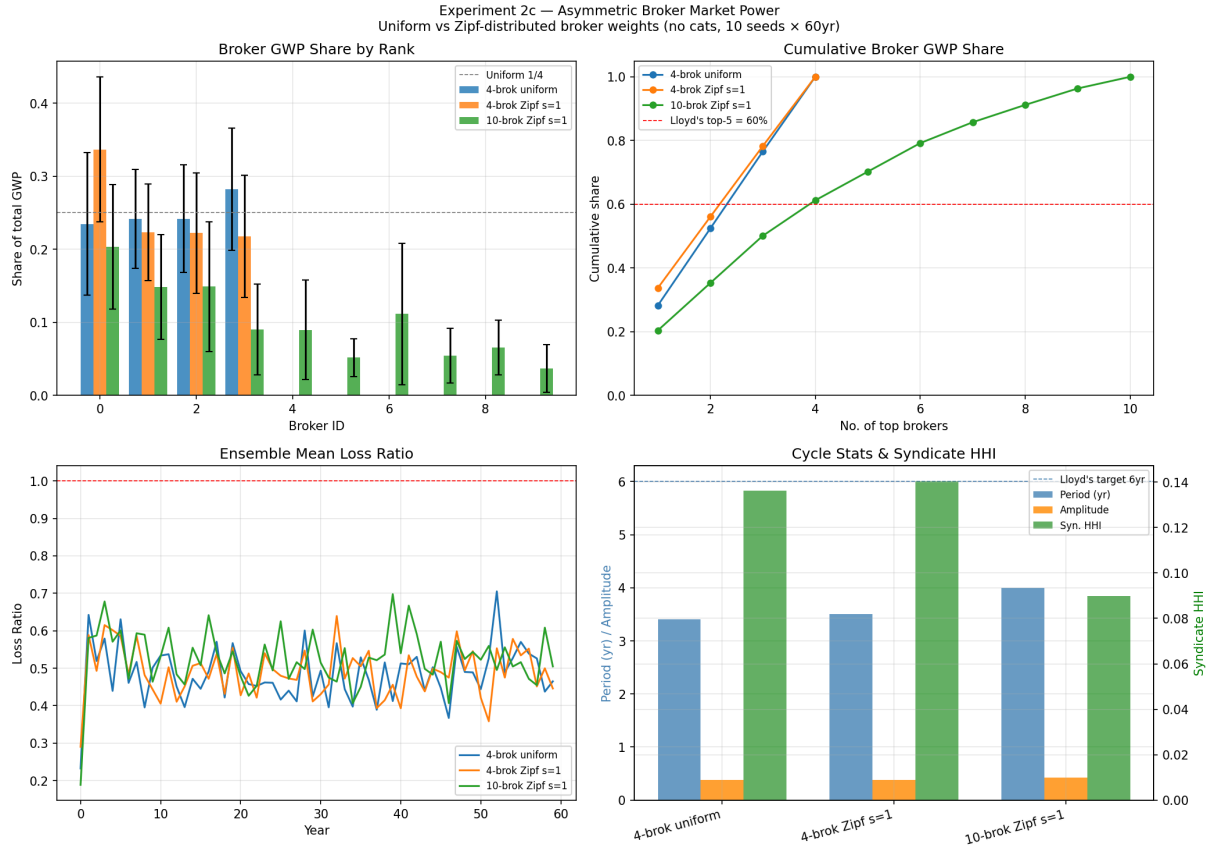


Figure 25: Broker market power sweep (Exp 2c): uniform vs Zipf routing, 4 vs 10 brokers. Cycle dynamics are insensitive to broker power.

6.4. Syndicate Market Entry (Exp 2d)

Entry rule: new syndicates enter when annual industry LR exceeds 100%. Mean starting capital equals the current market average.

Config	Period	Amplitude	Mean LR	HHI	Mean Entries
A — no entry (cf=0.06)	3.4 yr	1.379	0.918	0.337	0.0
B — entry (cf=0.06)	3.7 yr	1.938	0.907	0.306	12.0
C — entry (cf=0.03)	3.7 yr	1.489	0.773	0.219	8.3

Entry reduces HHI (0.337→0.219 with moderate cats), partially offsetting cat-driven insolvency concentration. With moderate catastrophe frequency (cf=0.03), entry also reduces mean LR (0.918→0.773): new entrants add capacity that holds prices down in the recovery phase. The mean entry rate of 12 entries over 60 years (0.2/yr) is consistent with Lloyd's admission of approximately one new syndicate per year in hard markets.

Two counter-intuitive findings: entry slightly lengthens cycle period (3.4→3.7 yr) as capital injections dampen the recovery overshoot; and amplitude rises with high cats + entry (1.379→1.938) because under-capitalised entrants fail in the next cat year, amplifying insolvency dynamics.

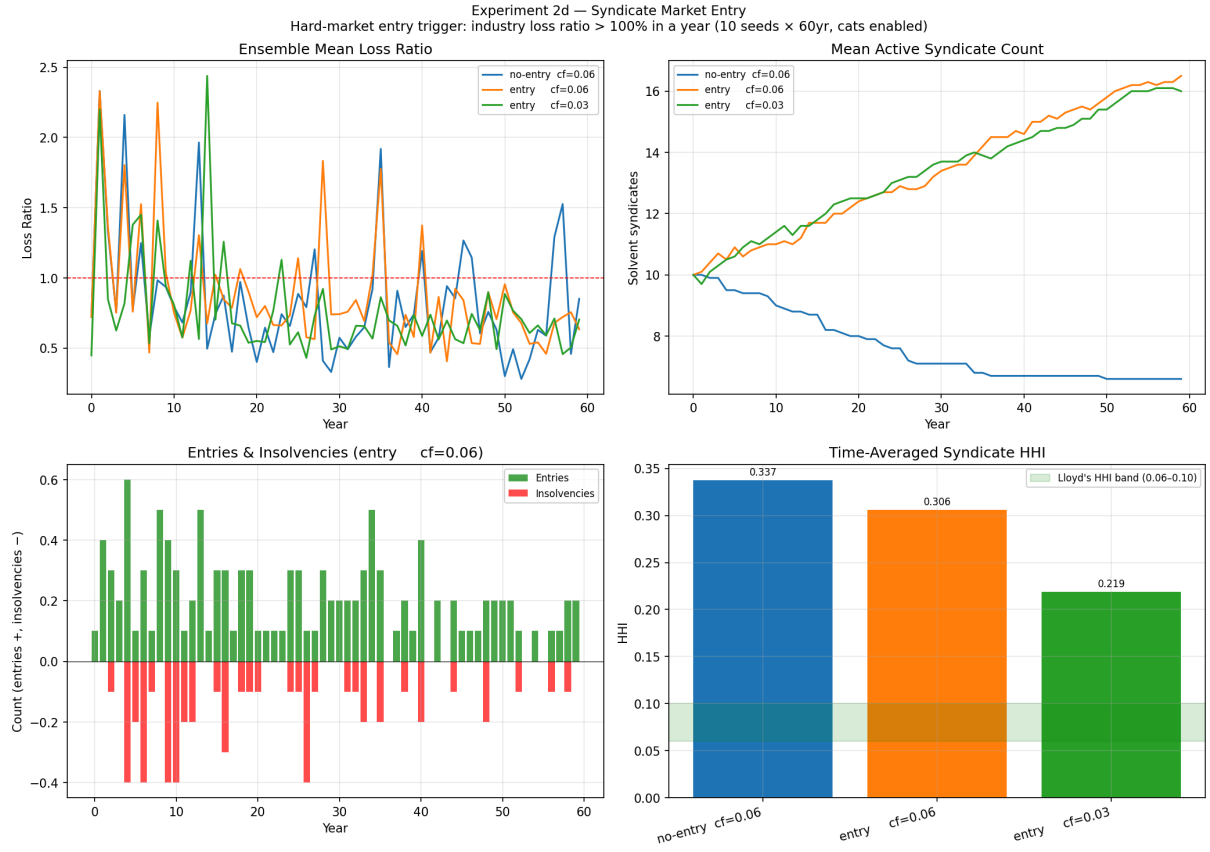


Figure 26: Syndicate market entry (Exp 2d): entry dynamics, HHI evolution, and new-entrant capital trajectories.

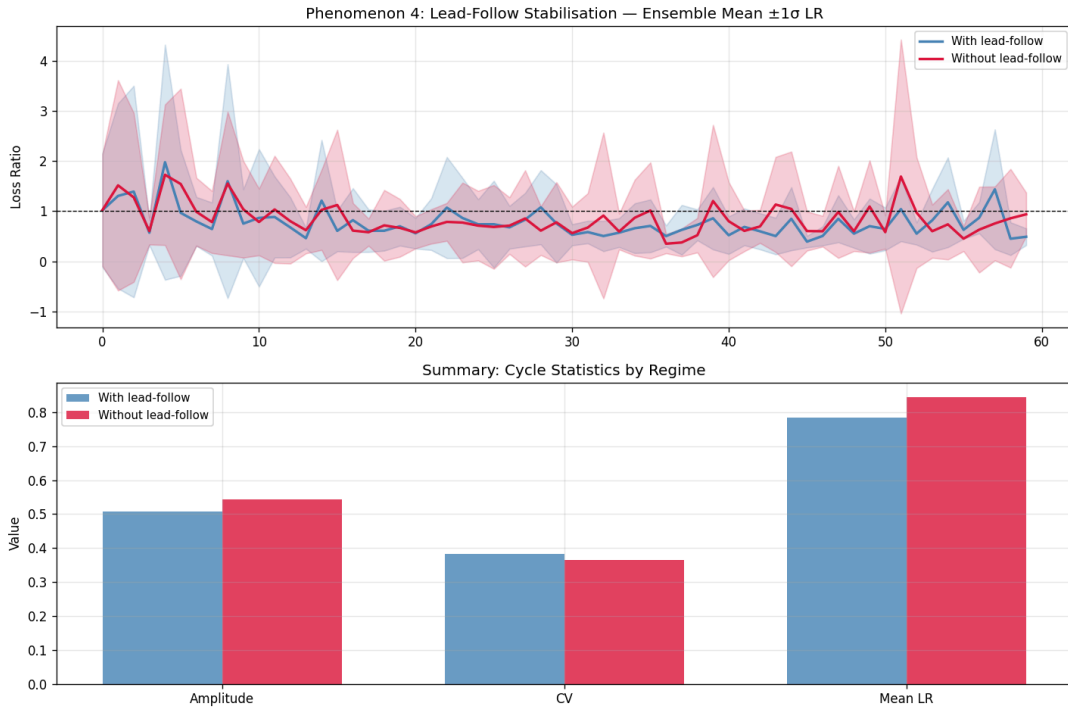


Figure 27: Lead-follow stabilisation: convergence of markup factors across syndicates over the simulation horizon.

7. Institutional Features (Tier 3)

7.1. Managed Runoff / Central Fund (Exp 3a)

When a syndicate becomes insolvent, its existing policies enter runoff: outstanding claims are paid by a Lloyd's Central Fund analogue. New risks are not written. This models the institutional mechanism by which Lloyd's protects policyholders from syndicate insolvency.

Config	Period	Amplitude	Mean LR	HHI	CF Total	CF%
A — no runoff (cf=0.06)	3.4 yr	1.379	0.918	0.337	0	0.0%
B — runoff	3.4 yr	1.463	0.937	0.337	1,419	1.7%
C — runoff + entry	3.5 yr	2.016	0.912	0.301	1,795	1.2%

The Central Fund covers approximately 1.7% of total claims in the catastrophe regime — a modest but meaningful backstop. Period and HHI are unchanged by runoff alone (structural market forces dominate), confirming that the runoff mechanism is an institutional safety valve rather than a market stabiliser. Reported LR is slightly higher with runoff because claims previously silently dropped at insolvency are now paid and recorded. The combination of entry and runoff reduces HHI further (0.337→0.301) while keeping CF cost manageable at 1.2% of claims.

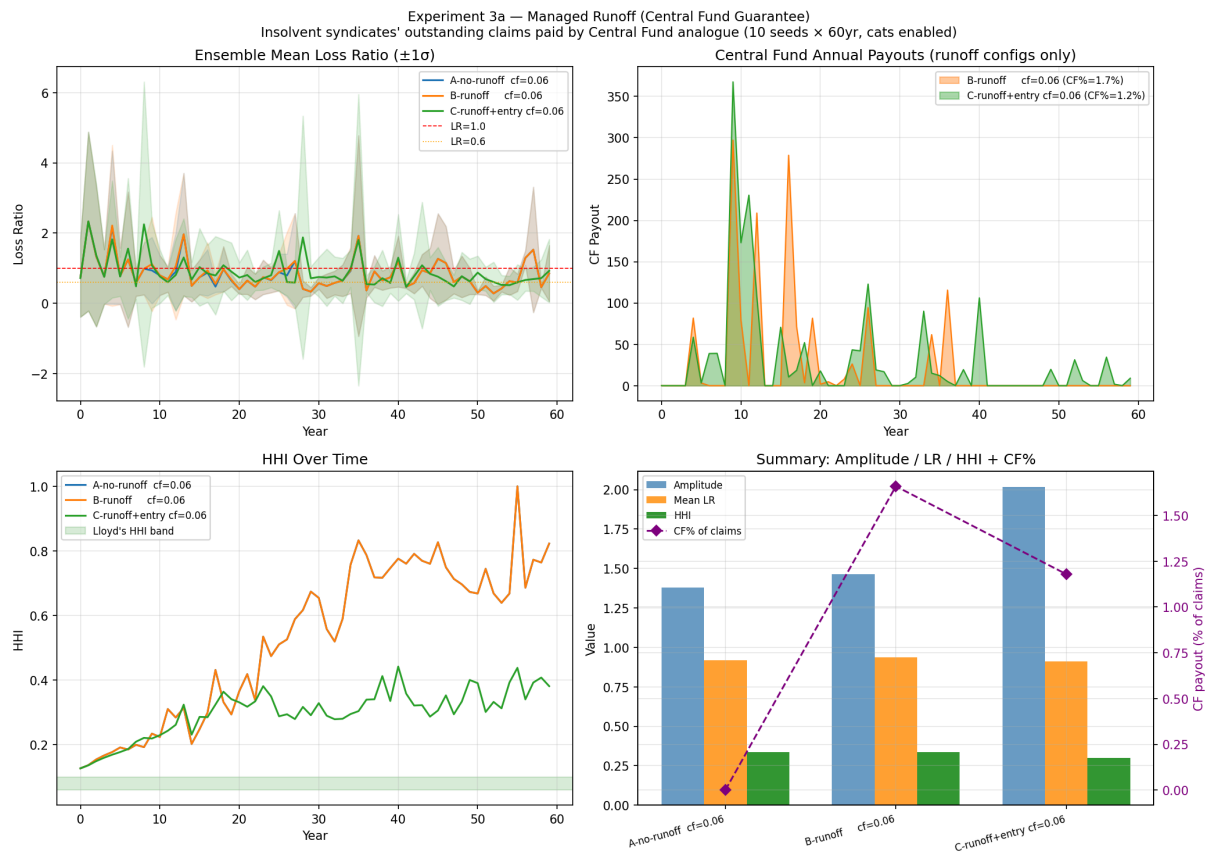


Figure 28: Managed runoff / Central Fund (Exp 3a): LR comparison with and without runoff, and cumulative Central Fund disbursements.

8. Cross-Experiment Synthesis: Key Market Phenomena

8.0.1. Phenomenon 1: The Underwriting Cycle is Structurally Robust

The cycle period of **3–4 years** is maintained across all nine parameter sensitivity sweeps and all nine structural experiments — agent count scaling, risk volume variation, line-of-business decomposition, per-region catastrophes, risk heterogeneity, broker power, market entry, and managed runoff. No intervention in any of these experiments altered the period by more than 0.6 years.

This robustness establishes that the cycle is structural, not parametric: it emerges from the β/w timescale mismatch inherent in the two-timescale pricing mechanism. The 2.7-year gap between the simulated period (3.3 yr) and the empirical Lloyd’s target (6 yr, Owadally 2018) is therefore also structural — it cannot be closed by any of the nine experimental mechanisms but requires direct modification of β and w in the model core.

8.0.2. Phenomenon 2: Catastrophes Are the Dominant Amplifier

Catastrophe frequency (`cat_freq`) is the single parameter with the largest effect on cycle amplitude, driving it from 0.38 (no cats) to 1.73 (`cat_freq`=0.12) — a 4.6-fold increase. Every structural addition in Tiers 1–3 (entry, runoff, broker power) leaves the cycle period unchanged but interacts with cat-driven insolvency concentration to modulate HHI and LR. Non-cat LOBs (Liability, Motor) are stable and profitable; cat LOBs (Property, Marine, Aviation) are persistently loss-making at current parameterisation.

Calibration implication: catastrophes are not optional for achieving realistic combined ratios. All no-cat configurations produce $CR \approx 0.82$, well below the 0.90–1.10 Lloyd’s target. Only the recommended+cats configuration achieves $CR = 1.019 \checkmark$.

8.0.3. Phenomenon 3: Market Concentration Responds to Catastrophe Shocks

HHI spikes after each cat-driven insolvency wave. The recommended no-cat configuration achieves $HHI = 0.070 \checkmark$ (within the Lloyd’s band), but enabling catastrophes pushes HHI to 0.218–0.392 as insolvencies concentrate the surviving market. Only syndicate market entry can partially restore competitive structure ($HHI\ 0.337 \rightarrow 0.219$ with moderate cats). This identifies market entry as the critical missing institutional mechanism for maintaining competitive structure in the presence of catastrophe shocks.

8.0.4. Phenomenon 4: The β/w Ratio is the Master Control

The cycle period is governed by the ratio of markup aggressiveness (β) to actuarial recency weight (w). High β/w produces a fast-responding, short-period cycle; low β/w produces a slow, long-period cycle. The 2D $\beta \times w$ heatmap confirms this: period varies along iso-ratio contours, not along individual parameter axes. All other parameters — capital, `cat_freq`, risk volume, agent count, LOB type, broker power — modulate amplitude and LR but leave period within the 3–4 year band. Reaching the empirical 6-year target requires reducing β/w to approximately half its current default value.

8.0.5. Phenomenon 5: Law of Large Numbers Dampens Cycles

Risk diversification progressively dampens cycle amplitude: 0.383 ($rpy=25$) $\rightarrow$ 0.257 ($rpy=50$) $\rightarrow$ 0.173 ($rpy=100$) along the proportional scaling sequence. More risks per syndicate reduce the idiosyncratic noise in each syndicate’s loss ratio, producing smoother aggregate market dynam-

ics and weaker oscillations. Crucially, this dampening requires proportional scaling (maintaining risks/syndicate constant): increasing agents without increasing risks produces capacity crowding and LR inflation, as demonstrated in Exp 0a. The recommended configuration ($n_{\text{syn}}=20$, $rpy=50$) achieves a balance between the law of large numbers effect and HHI calibration.

8.0.6. Phenomenon 6: Broker Network Effects Are Structurally Neutral

Routing concentration through dominant brokers (Zipf market power) does not alter market-wide pricing dynamics. Period, amplitude, and LR are essentially unchanged across all broker power configurations. Syndicates see the same risk types regardless of broker identity, because broker specialisms route risks by peril class — the aggregate risk mix presented to the market is unaffected by which broker channels that flow. Broker power affects market concentration metrics (who gets business) but not market dynamics (how prices evolve).

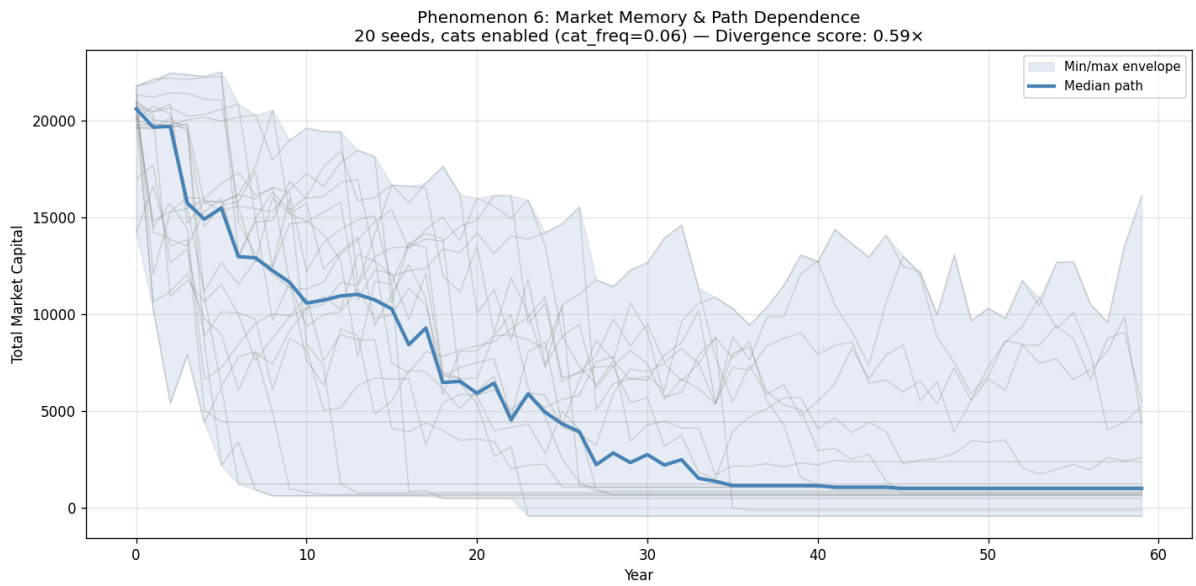


Figure 29: Path dependence: capital trajectories across 20 seeds. Stochastic variation in claim timing produces divergent long-run capital paths from identical initial conditions.

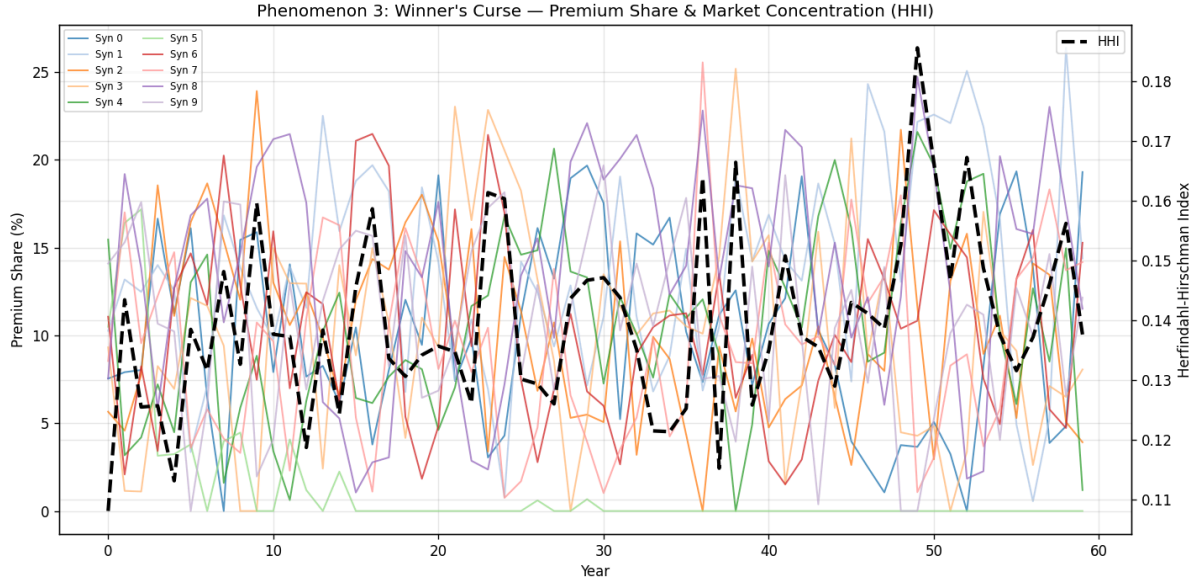


Figure 30: Market concentration (HHI) and premium share evolution over 60 years.

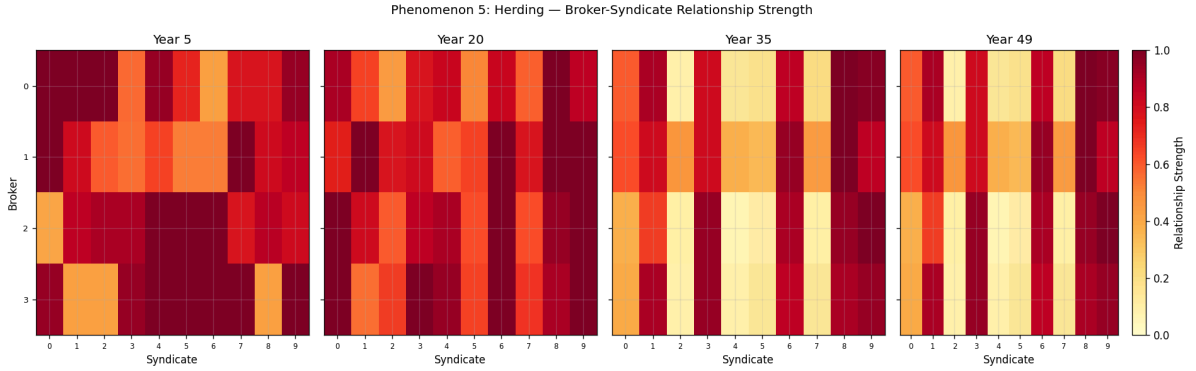


Figure 31: Broker-syndicate network evolution at years 5, 20, 35, and 49. Edge weight encodes cumulative placement volume; herding toward preferred syndicates emerges over time.

9. Calibration Assessment

Comparison of the recommended+cats configuration against all six Lloyd's benchmarks:

Benchmark	Lloyd's Target	Recommended + Cats	Status
Cycle period	5–7 yr	3.3 yr	✗ –2.7 yr short
HHI (syndicate GWP, no cats)	0.06–0.10	0.070	✓
Combined ratio	0.90–1.10	1.019	✓
Return on capital	5–15%	–17.6%	✗ structurally negative
Cat LR spike	+30–50 pp	+20.6 pp	✗ –9 pp short
Broker concentration (top 5)	60% of GWP	70.2% (10-brok Zipf)	≈ (slightly high)

What passes: Combined ratio (recommended+cats), HHI (recommended, no-cats), broker concentration (10-broker Zipf), market entry rate (0.2/yr $\approx$ Lloyd's 1/yr hard market rate).

What fails, and why:

- **Cycle period (−2.7 yr):** Structural gap from β/w ratio. Fix requires reducing β toward 0.15–0.25 and/or reducing w toward 0.05–0.10 to lengthen actuarial memory. Cannot be fixed by any of the nine experimental mechanisms.
- **Return on capital (structurally negative):** Premium/capital ratio in the simulation is approximately 0.02 — syndicates write only a tiny fraction of their capital as premium. Real Lloyd's syndicates write at approximately $1.5\times$ stamp capacity. The RoC metric requires the stamp capacity mechanism (Tier 3b) to be well-defined and calibrated.
- **Cat LR spike (−9 pp):** The current `cat_freq=0.05` produces a +20.6 pp spike in cat years. Increasing to 0.08–0.10 or raising catastrophe severity would close most of this gap, but would also further inflate mean LR and HHI.

10. Open Questions and Future Work

10.1. Immediate Calibration Priorities

1. **β/w recalibration to target 6-year period.** The β/w ratio should be reduced to approximately half the current default value. A grid search over $\beta \in [0.10, 0.25]$ and $w \in [0.05, 0.15]$ would identify the parameter region consistent with a 5–7 year period while maintaining other calibrated metrics.
2. **LOB_PARAMS recalibration.** The default LOB parameters (Motor $\lambda=2.0$, Liability $\lambda=1.5$) produce $LR \gg 1.0$ in Exp 2b. Recalibrating so that the frequency-weighted mean $\lambda \approx 0.6$ (the baseline) would restore realistic per-LOB LR without requiring changes to the actuarial pricing mechanism.
3. **Monetary rescaling sanity check (Exp 0c).** All dynamics are ratio-based (LR, markup factor, HHI). Rescaling monetary constants by 10,000 (`initial_capital = 20M`, risk limit `U(5M, 20M)`) should produce identical LR/HHI/period. Verifying this confirms the model is scale-invariant and monetary calibration is decoupled from dynamics.

10.2. Proposed Tier 3 Experiments

1. **Stamp capacity (Tier 3b).** Each syndicate's annual GWP is capped at a regulatory ceiling (stamp capacity) approved by Lloyd's Corporation, independent of current capital. This mechanism limits rapid soft-market expansion and is required for a well-calibrated RoC metric.
2. **Proportional reinsurance (Tier 3c).** Syndicates cede $X\%$ of each policy at a proportional premium, transferring catastrophe losses outside the market. Reinsurance is a key stabiliser in real Lloyd's; its absence may contribute to the over-violent cycles observed in cat regimes.

10.3. Modelling Improvements

1. **Full event-sourcing closure.** Emitting `DividendPaid` and `PremiumDistributed` events, adding `broker_id` to `PolicyBound` and `RiskDeclined`, and completing the `SyndicateInsolvent` payload would make the event store self-sufficient for all standard projections without restructuring the mutable-state architecture.

2. **Branching and counterfactual analysis.** Serialisable state snapshots would allow mid-simulation parameter changes and regulatory intervention experiments, isolating treatment effects from seed divergence.

11. References

Label	Citation
Wang 2017	Yunpeng, “Study on the Underwriting Circle of Property-Liability Insurance Market under Competition”
Owadally 2018	Owadally, Zhou & Wright, “The Insurance Industry as a Complex Social System: Competition, Cycles, and Crises”
Owadally 2019a	Owadally et al., “Time Series Data Mining with an Application to the Measurement of Underwriting Cycles”
Owadally 2019b	Owadally et al., “An agent-based system with temporal data mining for monitoring financial stability on insurance markets”
Olmez 2024	Olmez, Ahmed, Kam, Feng & Tua, “Exploring the Dynamics of the Specialty Insurance Market Using a Novel DES Framework: a Lloyd’s of London Case Study”
Axtell & Farmer 2022	“Agent-Based Modeling in Economics and Finance: Past, Present, and Future”

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